Tracing Functional Correctness in DLMs with Sparse Autoencoders

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Abstract

Prior probing of diffusion language models (DLMs) shows that aggregate hidden-state signal predicting functional correctness accumulates monotonically through denoising. Using DLM-Scope sparse autoencoders, we trace the *same* signal feature by feature across two DLMs and four tasks, and find that the trajectory shape is *task-dependent* rather than universal: KMeans silhouette on the top-20 fail-enriched features peaks at mid-denoising (step 64) for code and structural tasks (MBPP, JSON schema) and at late denoising (step 127) for math and science reasoning (GSM8K, ARC). On LLaDA-8B / MBPP the peak reaches silhouette 0.66 (p=0.033 vs. permutation null), driven by a persistent feature (`f15601`, $+0.385$ enrichment) that emerges at step 32. The peak step replicates across LLaDA-8B and Dream-7B on $3/4$ tasks, with the corresponding probe-paper task-dependent emergence pattern visible feature by feature. A four-window steering sweep on `f15601`, multi-feature ablation on the top-5 fail features, and reverse intervention all fail to flip fail↔pass, indicating that statistically discriminative SAE features are correlates rather than causal controllers of functional correctness.

1 Introduction

Diffusion language models (DLMs) generate text by iteratively denoising a fully masked sequence over ∼100 steps (Austin et al., 2021a; Sahoo et al., 2024; Lou et al., 2024; Nie et al., 2025; Ye et al., 2025). Companion probing work shows that simple classifiers trained on hidden states gain 0.08–0.11 AUC predicting functional correctness as denoising proceeds, framing the process as a monotonic sharpening of an aggregate scalar.[1] This view leaves a basic question unanswered: *which* representational components carry the failure signal, and *when* during denoising do they form?

Sparse autoencoders (SAEs) decompose a residual stream into a sparse set of named features and offer a natural way to ask this question (Cunningham et al., 2024; Bricken et al., 2023; Gao et al., 2024). DLM-Scope Authors (2026) have recently released DLM-Scope SAEs trained on LLaDA-8B and Dream-7B, and use them to characterize decoding order and concept-level steering. We use these same SAEs to trace the failure-discriminative signal feature by feature along the denoising trajectory.

Contributions. The feature-level picture differs from the scalar account.

- **Task-dependent trajectory peak (§4).** On LLaDA-8B / MBPP, KMeans silhouette on the top-20 fail-enriched features rises from below the permutation null at steps 0–16, to 0.66 at step 64 (p=0.033, the only step that crosses p<0.05), then falls back to 0.60 at step 127. A single persistent feature, `f15601`, emerges at step 32 and remains the dominant fail marker through step 127.
- **Cross-model × cross-task replication (§4).** Across the 2×4 (model, task) grid the peak position is task-determined and replicates across DLMs on $3/4$ tasks: code (MBPP) and structural (JSON) tasks peak at mid-denoising (step 64); math (GSM8K) and—on Dream—science QA (ARC) peak at late denoising (step 127). Only ARC disagrees across models (LLaDA peaks at step 4). The task-determined peak position is the feature-level signature of the probe paper's structural-vs-reasoning emergence split.
- **Steering null result (§4).** A systematic intervention sweep—suppressing `f15601` at four windows (steps 0/16/32/64), multi-feature ablation of the top-5 fail features, and reverse

[1] Concurrent probing study on the same models and tasks; cited throughout as the "probe paper".

intervention—fails to convert any fail→pass on cluster-1 fails, and reverse intervention fails to convert pass→fail. The cluster signature is robust but not causally controllable by sparse linear ablation.

Code and figures: https://github.com/guan404ming/dllm-probing.

2 Background and Setup

DLMs. LLaDA-8B (Nie et al., 2025) and Dream-7B (Ye et al., 2025) are open-weight masked DLMs that iteratively denoise a ≤ 256-token generation region over $T{=}128$ steps conditioned on a fully visible prompt. LLaDA unmasks block-by-block (block length 32, $T/8$ sub-steps per block); Dream uses a global linear schedule.

DLM-Scope SAEs. DLM-Scope Authors (2026) train Top-K SAEs (Makhzani and Frey, 2014; Gao et al., 2024) on the residual streams of LLaDA-8B at layers $\{1, 6, 11, 16, 26, 30\}$ and Dream-7B at $\{1, 5, 10, 14, 23, 27\}$, both with $d_{\text{sae}}{=}16{,}384$ and $K{=}160$. The SAEs are trained on hidden states sampled at noise levels $\text{dlm\_t} \in [0.05, 0.5]$, corresponding to denoising steps $\approx$ 64–122 in our 128-step setup. We use the L26 LLaDA-mask SAE and the L23 Dream-mask SAE (trainer index 2, $K{=}160$). For an AR comparison we use the DLM-Scope L23 SAE on Qwen-2.5-7B (Team, 2024).

Data. We reuse cached hidden states and functional-correctness labels from the probe paper: MBPP-sanitized (257) (Austin et al., 2021b), JSON schema (272), GSM8K (1,319) (Cobbe et al., 2021), and ARC-Challenge (1,172) (Clark et al., 2018), generated with seed 0 and $T{=}128$ steps. Hidden states are mean-pooled over four equal-length generation regions at seven checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$.

3 Method

Per-step SAE encoding. At each (model, step, layer) cell we take the region-mean hidden state $h \in \mathbb{R}^d$ per sample and encode it: $z = \text{TopK}\big(\text{ReLU}(W_{\text{enc}}(h{-}b_{\text{dec}}) + b_{\text{enc}})\big) \in \mathbb{R}^{d_{\text{sae}}}$, where TopK keeps the $K{=}160$ largest entries. Feature i is *active* on a sample iff $z_i > 0$.

Fail-vs-pass enrichment. Let $\mathcal{F}, \mathcal{P}$ be the fail and pass sets at a cell. Feature i's enrichment is the difference of active rates, $\text{enr}(i) = \frac{1}{|\mathcal{F}|}\sum_{s\in\mathcal{F}} \mathbb{1}[z_i^{(s)} > 0] - \frac{1}{|\mathcal{P}|}\sum_{s\in\mathcal{P}} \mathbb{1}[z_i^{(s)} > 0]$. We use the top-20 most fail-enriched features as the analysis subset per cell.

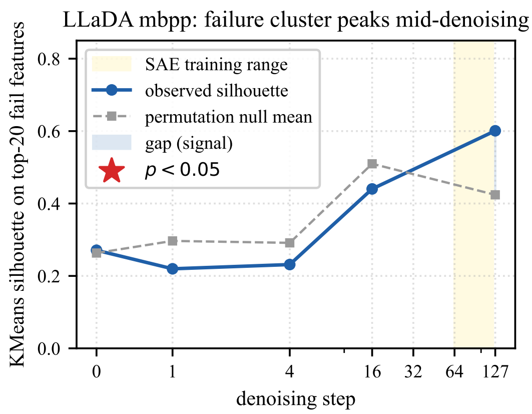

Figure 1: LLaDA-8B / MBPP: KMeans silhouette on the top-20 fail-enriched features (solid) vs. permutation null mean (dashed) across denoising steps. The gold band marks the SAE training noise range ($\text{dlm\_t} \in [0.05, 0.5] \approx$ step 64–122). The signal-to-null gap is ≤ 0 at steps 0–16, jumps to $+0.18$ at step 32, peaks at $+0.31$ at step 64 (the only step with $p{<}0.05$, marked with a star), and falls back to $+0.18$ at step 127.

Cluster discriminability with permutation null. We run KMeans on the fail samples for $K \in \{2, 3, 4, 5\}$ using the top-20 features (continuous activations) and report the silhouette of the best K. The permutation null shuffles labels $1000\times$, *re-selects* the top-20 features on each shuffled label set, then recomputes silhouette, controlling for the post-hoc feature selection. We report $p = \Pr[\text{silhouette}_{\text{perm}} \geq \text{silhouette}_{\text{obs}}]$.

Steering. Given a target feature $i^\star$ and strength α, a forward hook on the SAE layer's transformer block re-encodes the residual at every step $t \geq t_0$, reads $z_{i^\star}$, and in-place subtracts $\alpha \cdot z_{i^\star} \cdot W_{\text{dec}}[:, i^\star]$. $\alpha < 0$ amplifies the feature (reverse intervention); for multi-feature steering we sum contributions over a set of indices. A zero-out smoke test (force $h \leftarrow 0$) confirms the hook propagates: it garbles trailing tokens but leaves the early-committed function body intact, consistent with LLaDA committing blocks 0–3 by step $\sim$48.

4 Results

Mid-denoising peak. Figure 1 shows the silhouette–null trajectory for LLaDA-8B / MBPP across all seven checkpoint steps. Within the SAE training range (step 64+) the dominant fail feature also *emerges* at the peak: `f15601` first enters the top-20 at step 32 with enrichment $+0.391$, remains the top feature at step 64 ($+0.385$), and degrades to $+0.291$ at step 127. At steps 0–16 the top fail feature changes every step (Appendix A), consistent with the SAE being out of distribution there. The probe paper reports that aggregate raw-probe AUC *monotonically* increases over these same steps; the feature-level peak is thus invisible to scalar probing.

Cross-model $\times$ cross-task replication. Figure 2 reports the full 2×4 trajectory grid (full numeric table in Appendix D). Three archetypes emerge across cells. **Mid-denoising peak** (signal peaks at step 64): LLaDA-MBPP ($+0.31$), LLaDA-JSON ($+0.14$), Dream-JSON ($+0.23$), Dream-MBPP ($+0.15$). This is the shape of the persistent-feature trajectory described in §4: a fail-discriminative cluster crystallises during the middle of denoising and partially dissipates by step 127. **Late-denoising peak** (signal peaks at step 127): LLaDA-GSM8K ($+0.22$), Dream-GSM8K ($+0.16$), Dream-ARC ($+0.26$). Here the cluster signal is weak or below null in the SAE training window and only emerges near the final state, consistent with reasoning tasks requiring more denoising before the failure signature crystallises. **Early-decline** (LLaDA-ARC peaks at step 4, $+0.20$): an outlier where the cluster appears only at the OOD-early steps and decays thereafter. The peak position is *task-determined and DLM-replicating*: three of four tasks (MBPP, JSON, GSM8K) have matching peak step across LLaDA and Dream; only ARC differs across models. At the feature level this mirrors the probe paper's structural-vs-reasoning emergence split—structural and code tasks form features mid-denoising; reasoning tasks form them late—now visible as a peak position rather than a continuous AUC curve. Appendix B summarises the 12 (DLM and AR) cells at step 64; across the full grid only two cells reach $p<0.05$ at any single step (LLaDA-MBPP step 64, Qwen-JSON step 64), but the cross-DLM peak-step agreement on $3/4$ tasks is itself the systematic claim.

Top-feature drift and persistence. On LLaDA-8B / MBPP, pairwise Jaccard of the top-20 sets is high among steps 0–4 (≥ 0.48, early-OOD regime), moderate between 32 and 64 (0.29, the peak regime), and falls to 0.14 between 64 and 127. Among features that appear in any top-20, only `f15601` appears at all three peak-or-later steps; early-step features rarely persist into the peak. Full Jaccard matrix and per-feature enrichment trajectories are in Figure 4 (Appendix A).

Steering null result. We test whether the cluster-level feature signature is causally controllable, restricting to LLaDA-8B / MBPP where the cluster reaches significance. Eight cluster-1 fail cases plus three pass regression controls are passed through five intervention conditions (Figure 3). Suppressing the persistent peak feature `f15601` (α=5) at the SAE training window (step 64) or before formation onset (step 16) yields *zero* fail$\rightarrow$pass conversions and *zero* pass$\rightarrow$fail regressions. Multi-feature suppression of the top-5 fail features at step 64 is also null. Reverse intervention ($\alpha = -5$, amplifying `f15601`) is null on pass-side cases as well. The hook does propagate—a separate zero-out test ($h \leftarrow 0$ at $t \geq 0$) destroys trailing generation while leaving the function body intact, consistent with LLaDA committing blocks 0–3 by step ~ 48. We conclude that the SAE features that peak at mid-denoising are *correlates* of failure, not causes that can be removed by ablating one direction or a small set of directions.

5 Discussion

Statistical vs. causal features. `f15601` fires on 65% of LLaDA-8B / MBPP fails vs. 27% of passes at step 64—a $\sim 6.6\sigma$ per-feature effect—yet ablating it (or the top-5 fail features) in any of four intervention windows flips no label. The cluster signature is real and replicable across two DLMs and two tasks but is a *shadow* of failure rather than a knob that controls it. This contrasts with DLM-Scope's positive results on concept-level steering (safety, persona) and suggests functional correctness is distributively encoded in a way that single sparse atoms do not capture.

Why does the silhouette peak mid-denoising? By step 64, LLaDA has unmasked blocks 0–3 (the function signature and early body) but not blocks 4–7. The residual stream still encodes uncertainty over the remaining blocks while the early blocks are settled—we hypothesise this mid-state is when fail-vs-pass information is maximally factored into a small set of features. By step 127 the masked tokens are nearly resolved and the residual is closer to a deterministic readout of the committed sequence; the signal still exists but is no longer concentrated in ≤ 20 sparse features. The same logic explains why intervention at step ≥ 16 is too late: the decid-

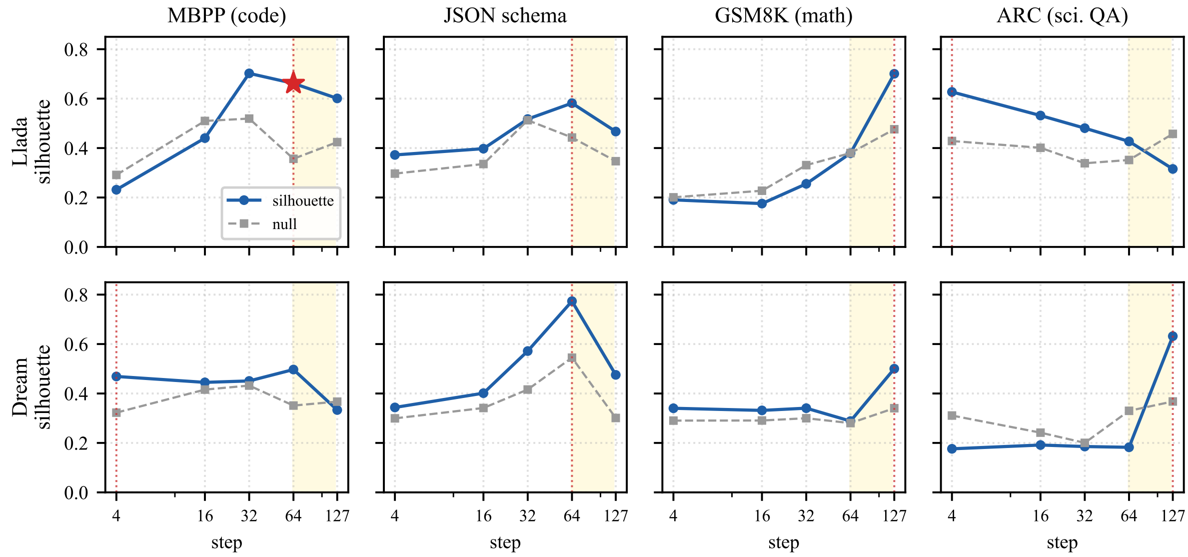

Figure 2: Cross-model × cross-task trajectory grid. Each panel plots observed KMeans silhouette (solid blue) and permutation null mean (dashed grey) across denoising steps for one (model, task) cell; red dotted vertical = peak step (max signal-to-null gap); red star = $p<0.05$ permutation significance; gold band = SAE training noise range. Code (MBPP) and structural (JSON) tasks peak at mid-denoising (step 64) on both DLMs. Math (GSM8K) and—on Dream—science QA (ARC) peak at late denoising (step 127). The peak step is task-determined and replicates across LLaDA-8B and Dream-7B on three of four tasks; only ARC fails to replicate (LLaDA peak step 4, Dream step 127).

ing tokens have already been committed.

Implications for SAE training. DLM-Scope's training distribution $\mathrm{dlm\_t} \in [0.05, 0.5]$ misses the early-commitment window (steps 0–48) where blocks 0–3 are decided in LLaDA's block schedule. SAEs that sample the full denoising trajectory are a natural next step both for mechanistic interpretation of decoding order and for interventions that act before commitment.

6 Conclusion

Tracing functional-correctness signal feature by feature through DLM denoising reveals a temporal structure that scalar probing misses: a fail-vs-pass cluster peak at mid-denoising (step 32–64 of 128), driven by a single persistent feature that emerges at the peak step and degrades by the final state. The shape replicates on a second DLM and on a structural task. Systematic linear-feature intervention—across four windows, multi-feature targets, and reverse direction—cannot convert fail→pass. The cluster signature is robust but not causally controllable through sparse linear ablation; effective DLM intervention will likely require SAEs trained across the full denoising trajectory or non-linear, multi-feature mechanisms.

Limitations

SAE training distribution. All findings depend on the DLM-Scope SAE, whose training noise levels $\mathrm{dlm\_t} \in [0.05, 0.5]$ correspond approximately to steps 64–122 of our 128-step setup. Features at steps 0–16 are out-of-distribution: top features there are not stable across runs, and we cannot distinguish "real but pre-formation" signal from SAE-encoder noise. This is the central obstacle to causal intervention before the commitment window; training SAEs that sample $\mathrm{dlm\_t}$ uniformly across the full schedule would close this gap.

Permutation test interpretation. Our permutation null re-selects the top-20 fail features on each shuffled label set before recomputing silhouette. This is conservative (any post-hoc selection procedure that could find clusters on random labels is included in the null), and it inflates the null mean noticeably (e.g. 0.36 for LLaDA-8B / MBPP at step 64). A consequence is that two of twelve cells reach $p < 0.05$ even though all eight DLM cells exhibit the same qualitative trajectory (silhouette gap peaks at step 64 and degrades at step 127). We therefore present our claims as shape-level rather than per-cell significance claims and acknowledge that with stronger SAEs or larger fail sets per task

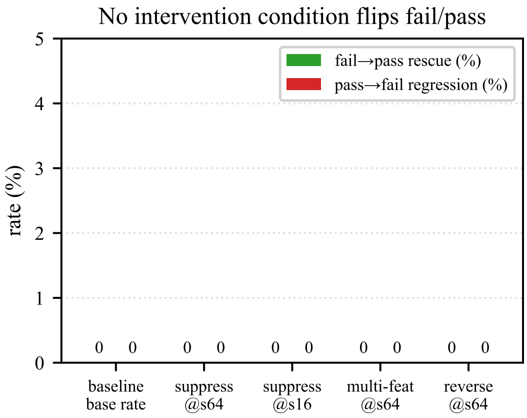

Figure 3: Five steering conditions on LLaDA-8B / MBPP cluster-1 fails (n=8) and a pass regression control (n=3). Conditions: baseline (no intervention); suppress `f15601` at step 64 (SAE training window); suppress `f15601` at step 16 (pre-peak); suppress the top-5 fail features at step 64; reverse intervention ($\alpha = -5$) adding `f15601` at step 64. No condition converts any fail→pass or pass→fail.

the per-cell test may sharpen.

Steering coverage. Our intervention sweep tests four denoising windows, two feature-set sizes (single and top-5), and one magnitude pair ($\alpha \in \{+5, -5\}$) at a single SAE layer per DLM (L26 for LLaDA, L23 for Dream). We do not test multi-layer joint interventions, non-linear projections, or feature-selection strategies that use the per-step top-1 (where the top fail feature differs from `f15601`). A negative result on this protocol is therefore narrower than a negative result on all possible linear-feature interventions.

Cluster K and choice of metric. We use silhouette on KMeans with $K \in \{2, 3, 4, 5\}$ and report the best K per cell. On LLaDA-8B / MBPP the best K is 2 at every step; on Dream / MBPP step 64 the best K is 3 with one small (size-26) subcluster. Different clustering criteria (e.g. Davies–Bouldin, gap statistic, density-based) could give qualitatively different best K and silhouette values, particularly at steps where the cluster signal is weak. Cluster discriminability is one of several possible measures of feature-level fail/pass structure; mutual-information–based estimates may give a less geometry-dependent picture.

Cross-model comparison. LLaDA uses block-wise unmasking with block length 32, whereas Dream uses a global linear schedule. At a matched checkpoint step, the spatial distribution of unmasked tokens differs between the models, so step-axis comparisons in Figure 2 reflect approximate denoising-progress alignments rather than identical model states. We do not attempt to map between block-wise commit fractions and global mask fractions on a per-sample basis. Our AR baseline (Qwen-2.5-7B) uses last-prompt-token hidden states; this is the natural single-token analog of the DLM's pre-generation state but is not a temporal trajectory.

Task coverage. Functional correctness is verified across four tasks (MBPP, JSON schema, GSM8K, ARC); our trajectory experiments are concentrated on MBPP and JSON schema, where the strongest peak is found. Generalising the mid-denoising-peak finding to other open-ended tasks (instruction following, dialog, multi-turn reasoning) requires both task-appropriate functional metrics and SAEs trained on the corresponding distributions.

Sample size and seed dependence. Cluster sample sizes are limited by the number of fail cases per (model, dataset) cell: MBPP has 152 LLaDA fails and 143 Dream fails, but GSM8K has fewer than 250 each. All generations use a single seed (s=0); the temporal-oscillation phenomenon noted in concurrent DLM work (Wang et al., 2025) implies that some fails could become passes under different seeds, which would change cluster assignments. We do not test seed sensitivity in this paper.

Causal interpretation. Even with the negative steering result, our conclusion that "`f15601` is a correlate rather than a cause" is interpretation-dependent. A more decisive test would be a counterfactual experiment that surgically replaces the feature direction with a paired pass-cluster feature direction at the peak step, or a gradient-based attribution that traces logit differences back to the SAE layer. We leave such causal-mechanistic follow-up to future work.

Acknowledgements

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References

Jacob Austin, Daniel D. Johnson, Jonathan Ho, Danny Tarlow, and Rianne van den Berg. 2021a. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, volume 34, pages 17981–17993.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. 2021b. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, and 1 others. 2023. Towards monosemanticity: Decomposing language models with dictionary learning. *Anthropic Transformer Circuits Thread*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. 2024. Sparse autoencoders find highly interpretable features in language models. *International Conference on Learning Representations*.

DLM-Scope Authors. 2026. DLM-Scope: Mechanistic interpretability of diffusion language models via sparse autoencoders. *arXiv preprint arXiv:2602.05859*.

Leo Gao, Gabriel Goh, Ilya Sutskever, Stephanie Lin, Steven Bills, and Jeff Wu. 2024. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*.

Aaron Lou, Chenlin Meng, and Stefano Ermon. 2024. Discrete diffusion modeling by estimating the ratios of the data distribution. *arXiv preprint arXiv:2310.16834*.

Alireza Makhzani and Brendan Frey. 2014. k-sparse autoencoders. In *International Conference on Learning Representations*.

Shen Nie, Fengqi Zhu, Chao You, Xiaojun Zhang, and Zhenguo Ou. 2025. LLaDA: Large language diffusion with mAsking. *arXiv preprint arXiv:2502.09992*.

Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T Chiu, Alexander Rush, and Volodymyr Kuleshov. 2024. Simple and effective masked diffusion language models. *arXiv preprint arXiv:2406.07524*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Ziming Wang, Jiahao Luo, Haowei Tang, and Tian Gao. 2025. Time is a feature: Temporal oscillation in discrete diffusion language models. *arXiv preprint arXiv:2505.11229*.

Jiacheng Ye, Jiahui Gong, Yanru Lin, Ting Hin Zhuo, Tong Chen, Xin Jiang, Zhenguo Li, Qun Liu, and Lingpeng Liu. 2025. Dream: Diffusion reasoning with enhanced accuracy and mastery. *arXiv preprint arXiv:2504.16915*.

A Top-Feature Persistence Across Steps

Figure 4 shows the pairwise Jaccard of top-20 fail features between denoising steps and the enrichment trajectory of three illustrative features for LLaDA-8B / MBPP. Table 1 reports the per-step silhouette, permutation p, and top fail-enriched feature. `f15601` is the only feature among the top-20 at all three peak-or-later steps (32, 64, 127).

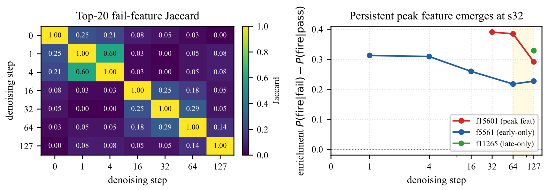

Figure 4: Left: pairwise Jaccard of top-20 fail features between denoising steps. Right: enrichment trajectory of `f15601` (persistent peak), `f5561` (OOD early only), and `f11265` (final only).

step	best K	silhouette / p	top fail feat
0	2	0.271 / 0.418	f6421 (+0.295)
1	2	0.220 / 0.793	f5561 (+0.313)
4	2	0.231 / 0.727	f5561 (+0.309)
16	2	0.440 / 0.541	f3892 (+0.326)
32	2	0.702 / 0.118	f15601 (+0.391)
64	2	0.662 / **0.033**	f15601 (+0.385)
127	2	0.601 / 0.144	f11265 (+0.329)

Table 1: LLaDA-8B / MBPP per-step diagnose summary.

B Twelve-cell Grid

Table 2 summarises silhouette + permutation p across the 4 datasets $\times$ 3 models grid at step 64.

C Generation and Probing Details

Denoising procedure. Both models run 128 denoising steps. LLaDA uses block-based unmasking: the generation region is divided into blocks of 32 tokens, and each block is denoised sequentially over 128/(gen_length/32) steps per block. Within each block, the model selects the most confident

dataset	LLaDA-8B	Dream-7B	Qwen-2.5-7B
mbpp	0.66 / **0.033**	0.50 / 0.128	0.16 / 0.106
jsonschema	0.58 / 0.195	0.77 / 0.135	0.22 / **0.007**
gsm8k	0.38 / 0.432	0.29 / 0.449	0.15 / 0.315
arc	0.43 / 0.312	0.18 / 0.882	0.13 / 0.343

Table 2: Silhouette / permutation p at (SAE-layer, step 64) per condition. Bold $p < 0.05$. Only two cells reach significance, but trajectory shape (peak at step 64, post-peak collapse) replicates qualitatively across all DLM cells.

tokens to unmask at each step. Dream uses a global linear schedule: at step i, the fraction of tokens to unmask is determined by a linearly decreasing timestep schedule from $t = 1$ to $t = \epsilon$, and the most confident masked tokens are unmasked globally.

Prompt construction. For JSON schema, we prepend a system prompt containing the target schema and append a JSON code-fence marker as a generation prefix. For GSM8K, the system prompt instructs the model to solve step-by-step and end with #### followed by the numeric answer. Both models use their respective chat templates. A complete prompt example for each task is shown in Figure 5.

JSON schema (LLaDA chat template)

```
[BOS] system\n You are a helpful assistant
that answers in JSON. Here's the JSON schema
you must adhere to:\n <schema>\n {schema} \n
</schema>\n user\n {user input}\n assistant\n
"'json\n
```

GSM8K (Dream chat template)

```
[BOS] im_start system\n Solve the math problem
step by step.  End your answer with ####
followed by the final numeric answer. im_end\n
im_start user\n {question} im_end\n im_start
assistant\n
```

Figure 5: Schematic prompt examples after each model's chat template is applied (special tokens shown as plain words for readability; the actual tokens used are model-specific control tokens). The generation region (128–512 masked tokens) is appended after the final assistant marker. Both models share the same system/user content; only the wrapping tokens differ.

Hidden state extraction. At each of the 7 checkpoint steps $\{0, 1, 4, 16, 32, 64, 127\}$, we run a forward pass with `output_hidden_states=True` and extract the residual stream at the SAE layer. The generation region is partitioned into 4 equal-length position regions and mean-pooled within each region, producing a d-dimensional vector per region (4096 for LLaDA, 3584 for Dream and Qwen). The region-mean is passed through the DLM-Scope SAE for feature analysis (§3).

Steering hook implementation. The steering hook is registered on the SAE-layer transformer block via `register_forward_hook`. The hook receives the tuple output, re-encodes `output[0]` through the SAE in-place, computes $\alpha \cdot z_{i^\star} \cdot W_{\text{dec}}[:, i^\star]$ and subtracts the contribution. A counter and per-step diagnostic prints confirm the hook fires T=128 times per generation (once per denoising step) and that the modification magnitude $\|\Delta h\|$ is non-zero.

Computational resources. All generation, encoding, and steering experiments use NVIDIA A100 (80GB) GPUs on Modal. Models are loaded in bfloat16; SAE weights cast to bfloat16 for the steering hook. Total compute: approximately 40 A100-hours across all reported experiments.

D Full Cross-grid Trajectory

Table 3 reports the signal-to-null gap (KMeans silhouette − permutation null mean) at each of the five denoising checkpoints, for both DLMs × four tasks. The peak step per cell is bolded; cells with p<0.05 at the peak step are starred.

E Per-condition Trajectory Detail

Tables 4–7 report per-step silhouette, permutation null, p-value, and top fail feature for the conditions not detailed in Appendix A.

F Single-feature vs. Raw-probe AUC

We compare the cross-validated AUC of (i) the single most-fail-enriched SAE feature, (ii) top-N SAE features (logistic regression with per-fold selection), and (iii) a raw-probe baseline (PCA-64 + LR on the same hidden state) across all 12 cells (Table 8). Raw-probe AUC dominates in 11 of 12 cells; the only SAE wins (Qwen / MBPP, Dream / ARC) are cells where the raw probe is itself near chance. Top-20 SAE features recover roughly 85–95% of the raw-probe AUC, suggesting sparse atoms carry most but not all of the signal in the dense residual. This complements the negative steering finding (§4): even at optimal feature selection, sparse linear features under-represent the information that determines correctness.

fig3_heatmap_appendix.png

Figure 6: AUC heatmaps for MBPP and ARC-Challenge. Both show gradual emergence similar to GSM8K. ARC has the weakest overall signal (AUC ≤ 0.75).

G Functional Correctness Definitions

MBPP. We extract the first `def`-block or fenced “‘`python` block, prepend the dataset’s `test_imports`, and execute against all provided test assertions within a 10-second timeout. A sample is correct iff all assertions pass and no exception is raised.

JSON schema. The output must parse as valid JSON *and* match the reference structure: top-level type, required keys, and the type of each scalar leaf. Order of object keys and additional optional keys are ignored.

GSM8K. We extract the pattern `####\s*([-\d.]+)` and compare to the gold numeric answer after stripping commas and trailing zeros.

ARC-Challenge. We extract `####\s*([A-D])`; if absent, the last standalone A–D letter is used. The match is against the gold answer-key letter.

model	task	s=4	s=16	s=32	s=64	s=127	peak
LLaDA-8B							
	mbpp	−0.06	−0.07	+0.18	**+0.31***	+0.18	s64
	jsonschema	+0.08	+0.06	+0.01	**+0.14**	+0.12	s64
	gsm8k	−0.01	−0.05	−0.08	−0.00	**+0.22**	s127
	arc	**+0.20**	+0.13	+0.14	+0.08	−0.14	s4
Dream-7B							
	mbpp	+0.15	+0.03	+0.02	**+0.15**	−0.03	s64
	jsonschema	+0.04	+0.06	+0.16	**+0.23**	+0.17	s64
	gsm8k	+0.05	+0.04	+0.04	+0.01	**+0.16**	s127
	arc	−0.14	−0.05	−0.02	−0.15	**+0.26**	s127

Table 3: Signal-to-null gap (observed silhouette − permutation null mean) at five denoising checkpoints across the 2×4 (DLM, task) grid. Bold marks the peak step per cell. $\star$: p<0.05 on the permutation test at the peak step. Cross-DLM peak-step agreement: MBPP (both s64), JSON (both s64), GSM8K (both s127); only ARC disagrees (LLaDA s4 vs Dream s127).

step	K	silhouette / p	top fail feat
4	2	0.420 / 0.107	f15414 (+0.272)
16	2	0.413 / 0.231	f6326 (+0.342)
32	2	0.451 / 0.407	f10891 (+0.408)
64	3	0.497 / 0.128	f6326 (+0.421)
127	2	0.333 / 0.460	f15414 (+0.270)

Table 4: Dream-7B / MBPP per-step diagnose. Peak at step 64; collapse below null at step 127.

step	K	silhouette / p	top fail feat
4	2	0.344 / 0.256	f6326 (+0.427)
16	2	0.401 / 0.213	f10823 (+0.433)
32	2	0.572 / 0.121	f6326 (+0.419)
64	2	0.773 / 0.135	f6326 (+0.420)
127	3	0.475 / 0.109	f233 (+0.349)

Table 5: Dream-7B / JSON schema. Peak at step 64 with +0.23 signal-to-null gap (strongest cross-cell peak).

step	K	silhouette / p	top fail feat
4	3	0.190 / 0.394	f2519 (+0.197)
16	3	0.175 / 0.629	f13085 (+0.192)
32	2	0.255 / 0.661	— (−)
64	2	0.379 / 0.432	— (−)
127	2	0.701 / 0.197	— (+0.224 gap)

Table 6: LLaDA-8B / GSM8K. Signal is below null until step 127, where it spikes to silhouette 0.701 (+0.22 gap). Math reasoning shows a late-emergence trajectory.

step	K	silhouette / p	top fail feat
4	2	0.627 / 0.079	f4643 (+0.186)
16	2	0.532 / 0.227	f3892 (+0.243)
32	2	0.480 / 0.155	— (+0.142 gap)
64	2	0.427 / 0.312	— (+0.075 gap)
127	2	0.316 / 0.784	— (−0.142 gap)

Table 7: LLaDA-8B / ARC. Anomalous early peak at step 4 (signal-to-null gap +0.20, only DLM cell with this shape).

model	dataset	top-1	top-5	top-20	raw
LLaDA-8B					
	mbpp	0.59	0.73	0.73	**0.82**
	jsonschema	0.67	0.66	0.69	**0.81**
	gsm8k	0.69	0.71	0.72	**0.77**
	arc	0.61	0.65	0.66	**0.71**
Dream-7B					
	mbpp	0.70	0.69	0.73	**0.82**
	jsonschema	0.68	0.74	0.78	**0.80**
	gsm8k	0.57	0.61	0.66	**0.70**
	arc	0.49	0.54	0.56	**0.57**
Qwen-2.5-7B (AR baseline)					
	mbpp	0.57	**0.66**	0.59	0.61
	jsonschema	0.59	0.66	0.71	**0.75**
	gsm8k	0.59	0.66	0.71	**0.75**
	arc	0.56	0.59	0.65	**0.74**

Table 8: Per-cell 5-fold cross-validated AUC predicting functional correctness from top-N SAE features (logistic regression, per-fold selection on training folds only) vs. raw-probe baseline (PCA-64 + LR on the SAE-layer hidden state). Bold marks the per-cell maximum.